

recall: decoding of HMM
trigram as example

$q(v/w, u)$ = prob of hidden state (tag v
given w, u before it

$e(x/v)$ = prob of word x occurring if the
tag at same pos is v

Decoding: given $x_1, \dots, x_n$

want to find $\max_{y_1, \dots, y_n} p(x_1, \dots, x_n, y_1, \dots, y_n)$

$$\begin{aligned} &= (\arg) \max_{y_1, \dots, y_n} p(y_1, \dots, y_n | x_1, \dots, x_n) \\ &= (\arg) \max_{y_1, \dots, y_n} \underset{\substack{\uparrow \\ q}}{p(y_1, \dots, y_n)} \underset{\substack{\uparrow \\ e}}{p(x_1, \dots, x_n | y_1, \dots, y_n)} \end{aligned}$$

to find $(\arg) \max$ idea: use DP

FD PP (i) make a table of intermediate sols/values
s.t. last entry/row is what you want

$$\begin{aligned} \pi(k, u, v) = \max_{y_1, \dots, y_k \in K^k} & p(x_1, \dots, x_k, y_1, \dots, y_k \text{ s.t. } y_k = v) \\ & \text{" } y_{k-1} = u \\ & \prod_{i=1}^k q(y_i | y_{i-2}, y_{i-1}) \times \prod_{i=1}^k e(x_i | y_i) \end{aligned}$$

$$\max_y p(x, y) = \max_{u, v} (\pi(n, u, v) \times q(\text{STOP} | u, v))$$

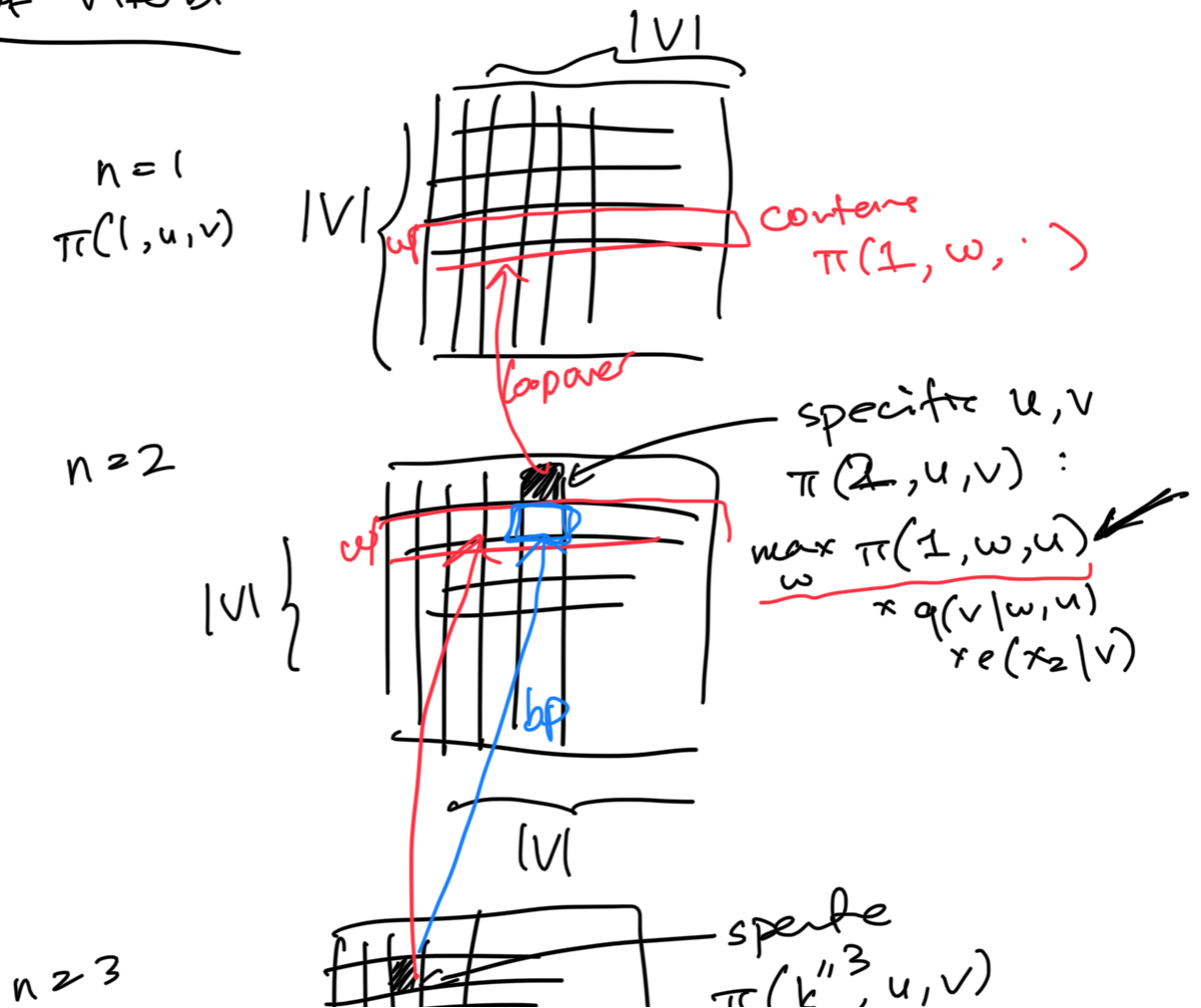
DP DP 2 values of table can be computed efficiently
(polynomial in your input sizes) from earlier
table values

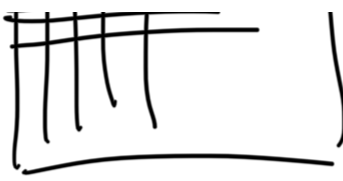
Prop $\forall k \in \{1, \dots, n\} \quad \forall u, v \in K$

$$\pi(k, u, v) = \max_{w \in K} \underbrace{\pi(k-1, w, u)}_{\substack{\text{takes into acct} \\ \text{all takes up to } y_{k-1} (=u) \\ + x_1, \dots, x_{k-1}}} \times q(v|w, u) \times e(x_k|v)$$

one missing $\nearrow$

Picture of Viterbi



$|K|$
 run time? $O(n \times |K|^2 \times |K|)$ $= O(n|K|^3)$
 look at / loop over all w $\pi(\frac{k-1}{2}, w, u)$
 cells in table of $\pi(k, \dots)$ for each cell loop over $|K|$ vals (row of prev. matrix) $\ll O(|K|^n)$

gives you $\max_{y_1, \dots, y_n} p(x_1, \dots, x_n, y_1, \dots, y_n)$

gives you prob of best fitting tags,
 but we really wanted the tags $y_1, \dots, y_n$ themselves

Fix? Viterbi w/ backpointers
 add a D.S.

when computing any cell

$$\pi(k, u, v) = \max_w \pi(k-1, w, u) \times \dots$$

also save the w that
 gave you max

$$bp(k, u, v) = \arg \max_w \pi(k-1, w, u) \times \dots$$

use this to read out max tag sequence

Adding bp, does it change run time?

doesn't change, just memoizing arg of arg max

$$O(n |K|^2)$$

In class questions

Q1 only possible tag seq

D N V STOP
 ↓ ↓ ↓
 the dog dog
 barks barks

Q2 Viterbi for 4-gram HMM

define π / its meaning

$$\pi(k, t, u, v) = \max \text{prob of } p(x_1, \dots, x_k, y_1, \dots, y_{k-2}, y_{k-1}, y_k)$$

$\begin{matrix} t & u & v \\ \uparrow & \uparrow & \uparrow \end{matrix}$

(English) max prob of subsequence
 $x_1, \dots, x_k$ w/ any tags that end
 in t, u, v

FP DP 1: solve final problem?

$$\max_{y_1, \dots, y_n} p(x_1, \dots, x_n, y_1, \dots, y_n) = \max_{t, u, v} \left(\pi(n, t, u, v) \times q(\text{STOP}(t, u, v)) \right)$$

FP DP 2: recursion, compute $\pi(k, t, u, v)$ using earlier table values

$\uparrow$
 a/b/c/d... u... v...

